

```

59  def main():
60      print("1. TCP Port Scanner")
61      print("2. Network Scanner")
62      choice = int(input("Enter your choice (1 or 2): "))
63
64      if choice == 1:
65          network = input("IP ADDRESS: ")
66          startPort = int(input("START PORT: "))
67          endPort = int(input("END PORT: "))
68          scanHost(network, startPort, endPort)
69      elif choice == 2:
70
71          target_ip = input("Enter the target IP address (e.g., 172.17.3.1/24): ")
72          network_scan(target_ip)
73      else:
74          print("Invalid choice. Please enter 1 or 2.")
75
76      if __name__ == "__main__":
77          main()
78          end = input("Press any key to close")

```

Figure 5: Defining main() function

```

1. TCP Port Scanner
2. Network Scanner
Enter your choice (1 or 2): 1
IP ADDRESS: 192.168.56.1
START PORT: 1
END PORT: 1000
[*] Starting TCP port scan on host 192.168.56.1
[+] 192.168.56.1:135/TCP Open
[+] 192.168.56.1:139/TCP Open
[+] 192.168.56.1:445/TCP Open
[+] TCP scan on host 192.168.56.1 complete
Press any key to close

```

Figure 6: Port scanning with range 1 to 1000

use of Python allows rapid development and easy integration with existing systems and tools, making the scanners accessible to a wide range of users.

Furthermore, the incorporation of libraries such as `socket` and `scapy` enhances the functionality and effectiveness of the scanners. These libraries provide comprehensive capabilities for network communication, packet manipulation, and analysis,

enabling thorough examination of network traffic and devices.

The open source nature of this project encourages collaboration and contribution from the cybersecurity

community. By sharing code and insights, developers can collectively improve and expand the capabilities of these scanners, ultimately benefiting the entire community. **END** 🐧

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```

1. TCP Port Scanner
2. Network Scanner
Enter your choice (1 or 2): 2
Enter the target IP address (e.g., 172.17.3.1/24): 172.16.133.1/24
Available devices in the network:
IP          MAC
172.16.133.2 7e:14:e2:14:8e:38
172.16.133.12 b4:8c:9d:08:c6:1b
172.16.133.17 74:f2:fe:27:ec:21
172.16.133.20 90:61:ae:28:5f:46
172.16.133.235 70:a8:d3:b8:05:33
172.16.133.25 70:bb:e9:1f:b2:5c
172.16.133.29 60:e9:ae:a2:91:fd
172.16.133.44 30:de:4b:1f:eb:ec
172.16.133.42 86:95:01:2b:45:34
172.16.133.38 1e:2f:68:23:ad:93
172.16.133.62 fc:01:7c:72:c8:01
172.16.133.40 5e:58:a2:e4:6c:66
172.16.133.76 ae:8f:6b:5a:86:39

```

Figure 7: Network scanning